

Science Olympiad
Circuit Lab B/C Test (**CLASS SET**)
UT Invitational



Written by: Darsh Shah, darshs1008@gmail.com

Directions:

- **This test is a class set.** Please do not write any answers on this paper.
- This was made using lastlegume's Scioly Test Template and Answer Sheet Generator. Thanks for the help! (If you're a test writer, I highly recommend checking it out: it's pretty cool)
- Write all answers on the answer sheets; any marks elsewhere will not be scored. If additional space is needed on a question, **clearly mark the question number and your team number on the top left of the page** and attach to your test.
- There is no penalty for wrong answers. Answer as many questions as possible, even if you aren't sure if you're correct. Additionally, try to answer parts of questions even if you are unable to answer the previous part: partial credit will be awarded for answers on the "right track".
- Points for each question are roughly indicative of intended question difficulty and of intended answer length (ie, easy questions with short answers are generally worth fewer points than difficult questions that require explanation).
- Units are required whenever giving a numerical answer. Don't worry about significant figures/decimal places unless the question states otherwise.
- Good luck on the test!

For grading use only

Question:	1	2	3	4	Total
Points:	63	31	27	11	132

Section A: Resistor Dividers as Sensing

Use your knowledge of circuits to answer the following questions. Partial credit will be awarded for partial responses to FRQs, especially for neat work that is close to the right answer.

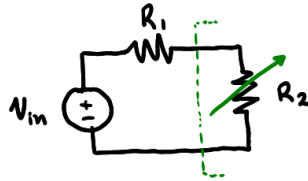


Figure 1: The image shows a generic resistor divider with R_1 as a fixed resistor and R_2 as a variable resistor. The green dashed line indicates physical separation between components.

1. (a) The following subparts refer to the circuit as it is drawn in the image. The voltages across the resistors are defined numerically (V_1 , V_2 for resistors R_1 , R_2 respectively)
 - i. (1 point) Write an equation that relates V_1 and V_2 to V_{in}
 - ii. (2 points) Write an equation for V_2 in terms of the resistances (R_1 , R_2) and input voltage (V_{in})
 - iii. (3 points) Sketch a graph for V_2 as a function of R_1 . Include the limiting cases (0 , ∞) and general shape.
 - iv. (3 points) Sketch a graph for V_2 as a function of R_2 . Include the limiting cases (0 , ∞) and general shape.
 - v. (3 points) Sketch a graph for V_2 as a function of V_{in} . Include the limiting cases (0 , ∞) and general shape.
 - vi. (2 points) Calculate the power dissipated by this resistor divider in terms of V_{in} , R_1 , and R_2 .
 - vii. (3 points) A common maximum value that microcontrollers can sense is 3.3V. Given that V_{in} is 5V, what ratio of R_1 to R_2 is best suited for to measure V_2 with this ADC? (Hint: ADC resolution is fixed by the number of bits and the voltage range it senses. Try ensure that V_2 fills the entire range)
 - viii. (1 point) Does this circuit for work AC input voltages? (Y/N)
 - ix. Using only the standard list of resistors below, make the following divider ratios using the fewest resistors possible resistors and the lowest DC currents (in that order).
 - $1k\Omega$, $3.3k\Omega$, $10k\Omega$, $22k\Omega$, $51k\Omega$, $100k\Omega$.
 - α) (1 point) $\approx 1 : 50$
 - β) (1 point) $\approx 1 : 100$
 - γ) (1 point) $\approx 1 : 10$
 - δ) (1 point) $\approx 1 : 3$
- (b) Consider the case where R_2 is a thermistor with a positive temperature coefficient (ie, the resistance of the device increases as the temperature increases). For this particular thermistor, assume that the temperature in Kelvin (T) and resistance(R) follow $R = AT + B$ where A and B are constants.
 - i. (2 points) What value would V_2 approach as the temperature increases?
 - ii. (2 points) What value would V_2 approach as the temperature decreases?
 - iii. (2 points) Write an expression for V_2 in terms of V_{in} , R_1 , and T .

- (c) Consider the case where R_2 is a potentiometer that is being varied mechanically such that its resistance is given by the figure below for the following subparts.

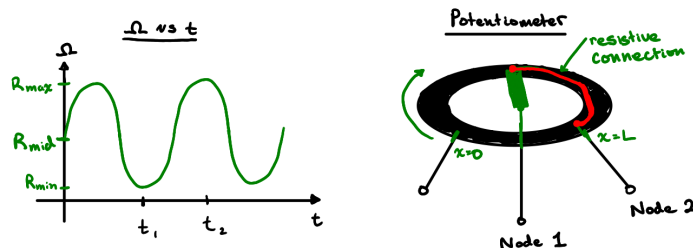


Figure 2: Left: Graph of resistance of potentiometer vs time. Right: Diagram of potentiometer internals. The resistance is set by the location of the middle wiper around a resistive loop. The position of the potentiometer is defined as $x = 0$ on the left end of the loop and $x = L$ at the right end. Note that the resistance of an object with length L , area A , and resistivity ρ is given by $R = \rho L/A$.

- i. (1 point) Assuming R_{min} is the smallest resistance between nodes 1 and 2, which location on the potentiometer corresponds to a resistance of R_{min} between nodes 1 and 2?
- ii. (1 point) Assuming R_{max} is the largest resistance between nodes 1 and 2, which location on the potentiometer corresponds to a resistance of R_{max} between nodes 1 and 2?
- iii. (3 points) In what wiper position is the potentiometer most likely to thermally fail assuming that voltage across the potentiometer is constant? (ie, when is the thermal density the highest?)
- iv. (3 points) Using the equation given in the caption of the figure, write an expression for the resistance of the potentiometer as a function of the position around the loop.
- v. (4 points) Graph the position of the wiper vs time that would be needed to generate the resistance vs time graph in the figure above.
- vi. The following subsubparts are about trying to use this circuit to generate AC across V_1 . Note that it is beneficial if the AC signal generated in this process has the largest amplitude possible.
 - α) (1 point) What does AC stand for?
 - β) (2 points) What must be true about the value of R_{min} compared to R_1 for the amplitude to be maximized?
 - γ) (2 points) What must be true about the value of R_{max} compared to R_1 for the amplitude to be maximized?
 - δ) (2 points) What is the period of V_1 with this changing resistance (in terms of t_1 and t_2)? What would this period have to be in order to generate an AC signal with the same frequency as the American grid?
 - ϵ) (3 points) Sketch the graph of V_1 for this changing resistance. Be sure to show how the shape is the same or different as the resistance graph.
 - ζ) (1 point) Is this a good AC source? (Y/N)
- vii. Generally, these sort of mechanical motions are generated using motors. This motor can be modeled as a rotating permanent magnet in a loop of wire.
 - α) (4 points) At what points in the magnet's rotation is the magnetic flux through the plane of the loop the largest? The smallest?
 - β) (2 points) Since the magnetic flux is changing, an opposing current is induced in the wires. What law defines this behavior?
 - γ) (3 points) If the number of poles in the permanent, rotating magnet are increased, what happens to the peak flux seen through the loop?
 - δ) (3 points) Can there ever be an odd number of poles present in the rotating magnet? (Y/N). Which of Maxwell's equations confirm this thinking?

Section B: PCB Planes and Capacitance

Use the parameters and your understanding of capacitance to answer the following questions.

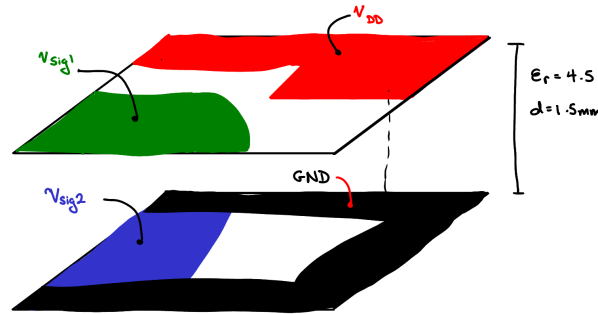


Figure 3: Diagram of a PCB with four relevant copper planes labeled v_{DD} , v_{sig1} , v_{sig2} , and GND on either the top and bottom layers of the PCB. Note that the material used to separate the layers has thickness (d) of 1.5mm and relative dielectric constant (ϵ_r) of 4.5 . The area of v_{sig1} with v_{sig2} beneath it is 20mm^2 . The area of v_{sig1} with GND beneath it is 10mm^2 . The area of v_{DD} with v_{sig2} beneath it is 2mm^2 . The area of v_{DD} with GND beneath it is 40mm^2 .

2. In calculations for capacitance, assume that there are no fringing fields between the planes.
 - (a) (2 points) What is the coupled capacitance between v_{sig1} and v_{sig2} ?
 - (b) (2 points) What is the coupled capacitance between v_{sig1} and GND?
 - (c) (2 points) What is the coupled capacitance between v_{DD} and GND?
 - (d) (2 points) Give an equality in terms of ϵ_r for two different materials that illustrates one reason the coupled capacitance between v_{sig2} and GND is very small.
 - (e) (1 point) Is v_{sig1} or v_{DD} able to carry more current?
 - (f) (3 points) Where, in relation to the planes, would there actually be large fringing fields in this figure?
 - (g) Assume that v_{sig1} is a clock signal (ie, a square wave with sharp edges that can be modeled as though v_{sig1} started at 0V and then quickly steps to v_{DD})
 - i. (1 point) Sketch the clock edge described in the statement above.
 - ii. (3 points) Given that any connection between two points using metal will have resistance, what type of circuit can model the shape of the clock edge including both the resistive and capacitive parameters?
 - iii. (4 points) Using your answer to the previous question (and assuming that the combined resistance seen by v_{sig1} is $50\text{m}\Omega$), what is the time constant to GND for this modeling circuit?
 - iv. (4 points) Given that a system generally stabilizes within 4τ , what is the maximum frequency that the clock can change at before the edges don't resolve before switching?
 - v. (2 points) Since v_{sig1} is a high speed node (ie, the voltage of the node changes quickly), there is current generated in v_{sig2} according to $i_c = C\Delta V/\Delta t$. Would this current be larger for a faster signal? (Y/N)
 - vi. (2 points) Is this resulting current a larger problem for the accuracy of v_{sig2} if v_{sig2} is large in amplitude or small in amplitude?
 - vii. (3 points) VDD is generally considered to be a constant supply voltage. Will the capacitance between VDD and GND generate any currents as in the previous subparts? Write an equation to support your answer.

Section C: Magic Boxes and Magistrates

Use your understanding of circuits, historical discoveries, and general knowledge to answer the following questions.

3. Use the figure below of a magic box to answer the following parts.

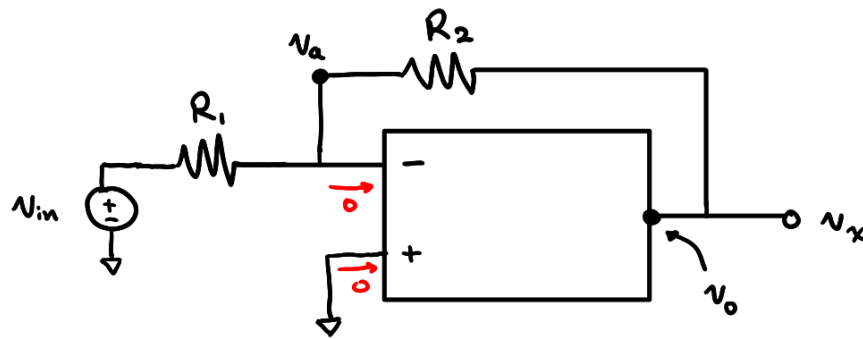


Figure 4: In the figure above, the magic box has a couple of properties. 1) the inputs to the magic box (the + and - marked inputs) are high resistance (ie, no current flows into the inputs). 2) the output of the magic box labeled v_o is given by $v_o = A(v_+ - v_-)$ where A is the gain of the box (a constant in this case). Note that the triangles are grounds (ie, $0V$).

- (2 points) Use Kirchhoff's Voltage Law (KVL) to express an equation for the voltages in this circuit (v_{in} , v_a , and v_x). Express each term in the loop explicitly.
 - (3 points) Write the current balance equation for node v_a using Kirchhoff's Current Law (KCL) in terms of voltages only (v_{in} , v_a , and v_x).
 - (2 points) Write v_x as a function of the voltages (v_{in} , v_a , and v_x).
 - (2 points) Combine (but don't simplify yet) the two equations above to get a single equation with only v_a , v_{in} , and system parameters (A , R_1 , R_2).
 - (4 points) Solve the equation above for v_a in terms of v_{in} and system parameters (A , R_1 , R_2).
 - (3 points) For very large gains (ie, $A \rightarrow \infty$), what is v_a ?
 - (4 points) Write and solve an equation for v_o in terms of v_{in} and system parameters (A , R_1 , R_2).
 - (3 points) For very large gains (ie, $A \rightarrow \infty$), what is v_o ?
 - (4 points) Using your understanding of this box, what would be a condition on the input voltages (v_+ and v_-) that could replace using the limit on A ?
4. Answer the following statements with the first letter of the scientist(s) from the rules ((A)mpere, (C)oulomb, (K)irchhoff, (V)olta, (O)hm, (T)esla, (F)araday) that satisfy the statement.
- (1 point) The law directly named after them has a $1/r^2$ dependence.
 - (2 points) The law directly named after them has a $1/r$ dependence.
 - (5 points) The law named directly after them is useful in steady state (ie, has no time dependence).
 - (3 points) They did most of their work in France.